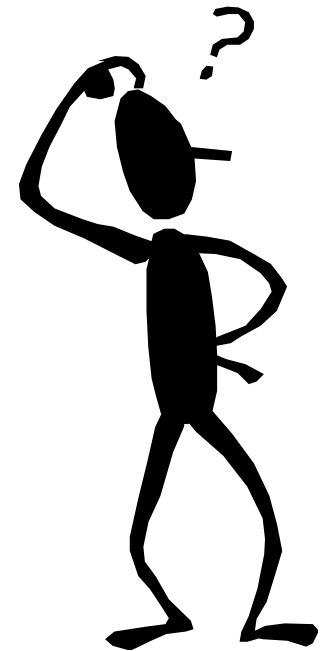


Design, prototyping and construction



Overview

- Prototyping and construction
- Conceptual design
- Physical design
- Generating prototypes
- Tool support



Prototyping and construction

- What is a prototype?
- Why prototype?
- Different kinds of prototyping
 - low fidelity
 - high fidelity
- Compromises in prototyping
 - vertical
 - horizontal
- Construction

What is a prototype?

In other design fields a prototype is a small-scale model:

- a miniature car
- a miniature building or town

What is a prototype?

In interaction design it can be (among other things):

- a series of screen sketches
- a storyboard, i.e. a cartoon-like series of scenes
- a Powerpoint slide show
- a video simulating the use of a system
- a lump of wood (e.g. PalmPilot)
- a cardboard mock-up
- a piece of software with limited functionality written in the target language or in another language

Why prototype?

- Evaluation and feedback are central to interaction design
- Stakeholders can see, hold, interact with a prototype more easily than a document or a drawing
- Team members can communicate effectively
- You can test out ideas for yourself
- It encourages reflection: very important aspect of design
- Prototypes answer questions, and support designers in choosing between alternatives

What to prototype?

- Technical issues
- Work flow, task design
- Screen layouts and information display
- Difficult, controversial, critical areas

Low-fidelity Prototyping

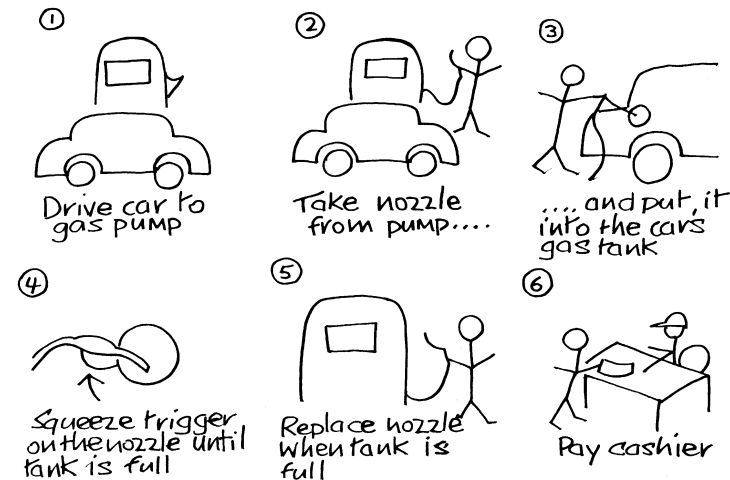
- Uses a medium which is unlike the final medium, e.g. paper, cardboard
- Is quick, cheap and easily changed
- Examples:
 - sketches of screens, task sequences, etc
 - 'Post-it' notes
 - storyboards
 - 'Wizard-of-Oz'

Storyboards

- Often used with scenarios, bringing more detail, and a chance to role play
- It is a series of sketches showing how a user might progress through a task using the device
- Used early in design

Sketching

- Sketching is important to low-fidelity prototyping
- Don't be inhibited about drawing ability. Practice simple symbols



Card-based prototypes

- Index cards (3 X 5 inches)
- Each card represents one screen or part of screen
- Often used in website development

Travel
Organiser

23 August 2006

WELCOME HELEN

Where do you want to go?

What date do you want to travel?

Which form of transport do you want?

Do you need accommodation?

Travel
Organiser

23 August 2006

Train timetable from Milton Keynes Central
to York
on 16.09.06

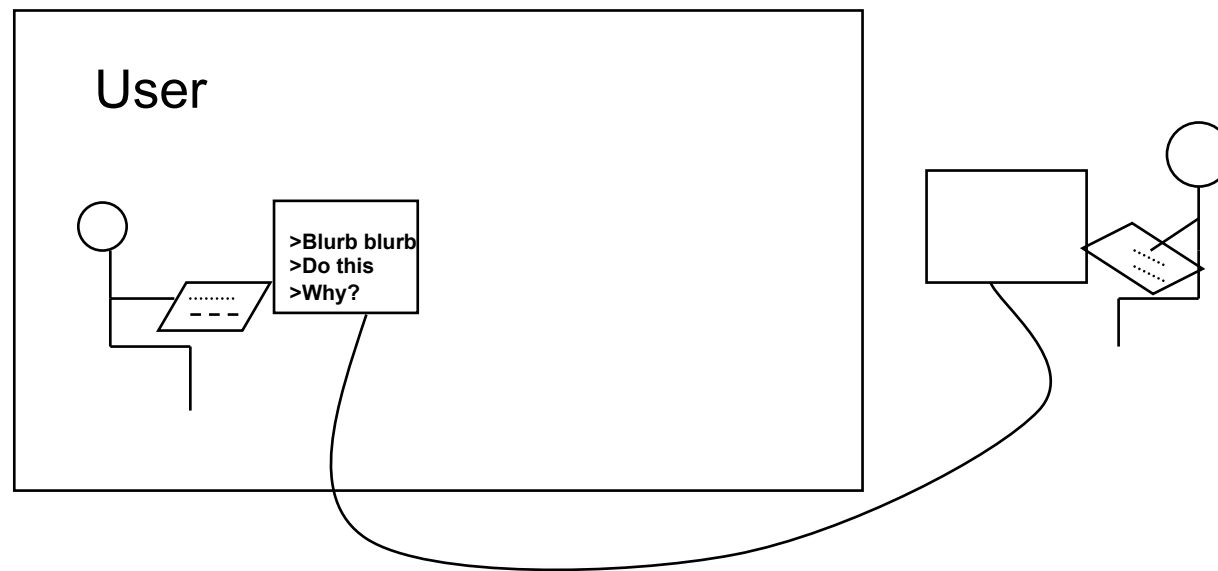
Depart	09:09	10:09	same	22:09
Arrive	12:30	13:30	mins past hour	01:30

Accommodation

Hotel	B&B
£40 to £150	£20 to £60

‘Wizard-of-Oz’ prototyping

- The user thinks they are interacting with a computer, but a developer is responding to output rather than the system.
- Usually done early in design to understand users’ expectations
- What is ‘wrong’ with this approach?



High-fidelity prototyping

- Uses materials that you would expect to be in the final product.
- Prototype looks more like the final system than a low-fidelity version.
- For a high-fidelity software prototype common environments include Macromedia Director, Visual Basic, and Smalltalk.
- Danger that users think they have a full system.....see compromises

Compromises in prototyping

- All prototypes involve compromises
- For software-based prototyping maybe there is a slow response? sketchy icons? limited functionality?
- Two common types of compromise
 - 'horizontal': provide a wide range of functions, but with little detail
 - 'vertical': provide a lot of detail for only a few functions
- Compromises in prototypes mustn't be ignored.
Product needs engineering

Construction

- Taking the prototypes (or learning from them) and creating a whole
- Quality must be attended to: usability (of course), reliability, robustness, maintainability, integrity, portability, efficiency, etc
- Product must be engineered
 - Evolutionary prototyping
 - 'Throw-away' prototyping

Conceptual design: from requirements to design

- Transform user requirements/needs into a conceptual model
- “a description of the proposed system in terms of a set of integrated ideas and concepts about what it should do, behave and look like, that will be understandable by the users in the manner intended”
- Don’t move to a solution too quickly. Iterate, iterate, iterate
- Consider alternatives: prototyping helps

Is there a suitable metaphor?

- Interface metaphors combine familiar knowledge with new knowledge in a way that will help the user understand the product.
- Three steps: understand functionality, identify potential problem areas, generate metaphors
- Evaluate metaphors:
 - How much structure does it provide?
 - How much is relevant to the problem?
 - Is it easy to represent?
 - Will the audience understand it?
 - How extensible is it?

Considering interaction types

- Which interaction type?
 - How the user invokes actions
 - Instructing, conversing, manipulating or exploring
- Do different interface types provide insight?
 - WIMP, shareable, augmented reality, etc

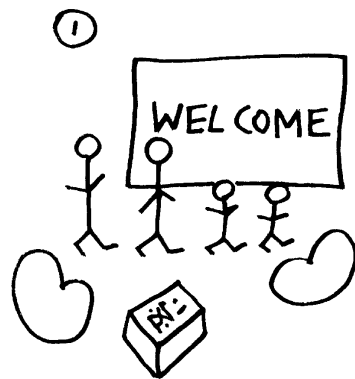
Expanding the conceptual model

- What functions will the product perform?
What will the product do and what will the human do (task allocation)?
- How are the functions related to each other?
Sequential or parallel?
Categorisations, e.g. all actions related to telephone memory storage
- What information needs to be available?
What data is required to perform the task?
How is this data to be transformed by the system?

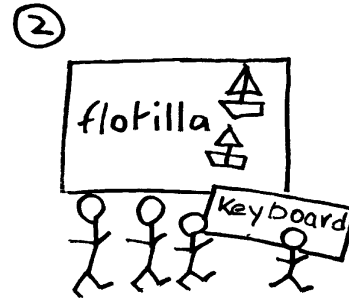
Using scenarios in conceptual design

- Express proposed or imagined situations
- Used throughout design in various ways
 - scripts for user evaluation of prototypes
 - concrete examples of tasks
 - as a means of co-operation across professional boundaries
- Plus and minus scenarios to explore extreme cases

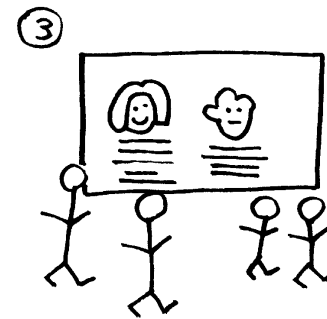
Generate storyboard from scenario



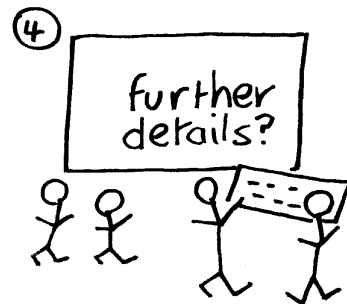
Thomson family
gather around



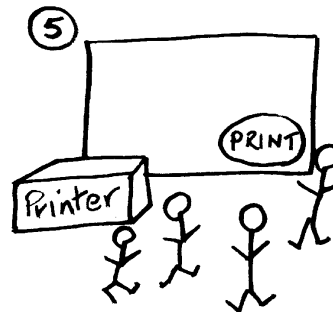
System suggests
flotilla



System shows
descriptions



System asks
for details



Summary printed

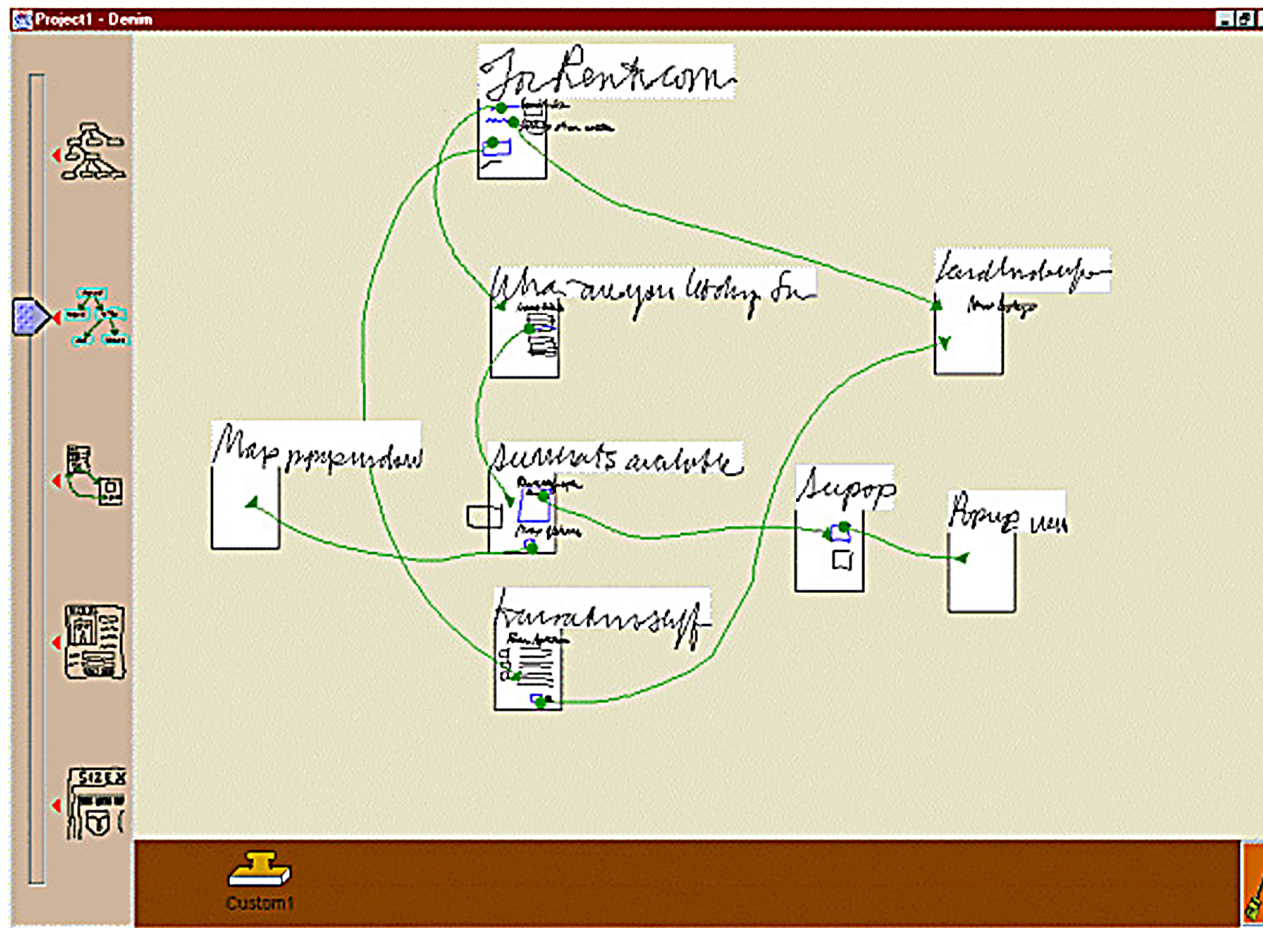
Generate card-based prototype from use case

TRAVEL INFORMATION
Visarequirements
Vaccination Recommendations
What to pack before you go

VISA REQUIREMENTS
Destination Country
Traveller's Nationality
Find Requirements

VISA REQUIREMENTS FOR (COUNTRY)
Print

Tool support - DENIM



Summary

- Different kinds of prototyping are used for different purposes and at different stages
- Prototypes answer questions, so prototype appropriately
- Construction: the final product must be engineered appropriately
- Conceptual design (the first step of design)
- Consider interaction types and interface types to prompt creativity
- Storyboards can be generated from scenarios
- Card-based prototypes can be generated from use cases